Journal of the Knowledge Economy (2024) 15:917–935
https://doi.org/10.1007/s13132-023-01108-z

Trade Diversion and Creation Effect of Free Trade Agreements in ASEAN: Do Institutions Matter?

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Received: 17 March 2022 / Accepted: 6 January 2023 / Published online: 21 January 2023
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Abstract
There has been a proliferation of free trade agreements (FTAs) over the last three decades and the Association of Southeast Asian Nations (ASEAN) has since greatly embraced the development. However, research on the effect of trade agreements on trade in the Southeast Asia region has not received adequate attention. The few studies conducted on the issue fail to account for the heterogeneity of the FTA effects and the role of the quality of domestic economic and political institutions in shaping the effects. Therefore, this study evaluates the role of institutions vis-a-vis the trade diversion and creation effects of the FTAs on trade flow in ASEAN over the period 2000–2018. The study accounts for FTA endogeneity and multilateral resistance terms by estimating a structural gravity model using Poison pseudo-maximum likelihood with a high-dimensional fixed effect (PPMLHDFE). The findings reveal that the inclusion of the indicators of political and economic institutions into the specification significantly changed the magnitudes of the trade creation and diversion effects of FTA on international and intra-ASEAN trade. And, the effects of the quality of institutions are robust to the different trade flows (total trade, exports, and imports) of the ASEAN economies. Therefore, we conclude that the trade diversion and creation effects of the FTAs on trade flows in ASEAN largely depend on the quality of economic and political institutions in the region.

Keywords Trade creation · Trade diversion · Institutions · Trade agreement

JEL Classification F13 · F15 · O24 · O43

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Introduction

The surge of trade agreements in the world over the last three decades is unprecedented. The World Trade Organization reported that over 300 trade agreements are in force as of June 2019. This is in a bid to facilitate the integration of the world market. The Southeast Asian countries have significantly keyed into the world integration drive via the engagement in bilateral and multilateral preferential free trade agreements with countries and regions across the globe. The increasing wave of the integration of the global market and its attendant effects on bilateral trade flows have attracted the attention of researchers and policymakers over the years. Studies have examined the welfare effects and the trade diversion or creation effects of the trade agreements on members and non-members respectively. Meanwhile, the discussion vis-à-vis the effect of the trade agreement on members' and non-members' trade remains contentious.

A near consensus has been reached that free trade agreements (FTA) promote bilateral trade among members. However, it is established that the facilitation of trade between members is often at the detriment of the trade with non-members (Lee & Shin, 2006). The preferential free trade agreements (FTA) could have an infinitesimal or negative effect on the trade flows. The agreements induce both trade creation and trade diversion which makes the effect of the agreements ambiguous (Viner, 1950). Trade creation refers to the substitution of a lower-cost source of supply for a higher-cost source of supply. On the other hand, trade diversion is the substitution of a more costly source of supply within the trade area for a less costly source outside the area. The harm from trade diversion could outweigh the gain of free trade agreements due to a reduction in terms of trade with non-members (Anderson & Yotov, 2016). This happens mostly when the members are involved in multiple overlapping FTAs with complex rules of origin (RoO) and copious customs provisions, which increase the transaction cost (Schüle & Kleisinger, 2016). Put differently, multiple FTA could slow down trade between member countries, a phenomenon known as the "spaghetti (noodle) bowl effect" (Bhagwati, 1995). The effect depends largely on the cost structures (including the quality of institutions) of the member and non-member countries.

Nonetheless, research on the effect of trade agreements on trade in the Southeast Asia region has not received adequate attention. The region is central to the integration of the world market. The Association of Southeast Asian Nations (ASEAN), formed in 1967, signed the oldest regional trade agreement in Asia, and currently, numerous trade agreements are in force in the region. For instance, in 2016, the Philippines, Thailand, and Malaysia are engaged in over 30 different trade agreements; Brunei Darussalam, Cambodia, and Indonesia have signed over 40 trade agreements while Vietnam and Singapore are involved in 51 and 72 trade agreements respectively (World Trade Organization (WTO), 2019). It is therefore imperative to evaluate the effects of the trade agreements on intra-regional, international, and total bilateral trade in the region. The few studies conducted on the issue in the ASEAN region are marred by numerous defects.

Although most of the studies have recognized the importance of the trade creation and diversion of the regional trade agreements, the relevance of the quality of institutions (economic and political) in facilitating or hindering trade creation and/or diversion has not been incorporated into the thesis. Political and economic institutions are essential determinants of trade flows (Alhassan & Payaslioğlu, 2019). Good institutions boost comparative advantage, reduce uncertainty and cost of transaction, and consequently facilitate trade while bad institutions lead to comparative disadvantage, raise the cost of trade, and thus hinder the flow of trade (Belloc, 2006; Zeynalov, 2017; Soeng & Cuyvers, 2018; Aziz et al., 2018; Osabuohien et al., 2018). Since, the trade creation and diversion effects depend on cost structure, ignoring institutions in the analysis could lead to misleading conclusions. Therefore, accounting for the domestic institutions in the analysis of trade diversion and the creative effects of trade agreements is indispensable.

This study contributes to the debate in the following folds. First, the study evaluates the trade diversion and creation effects of FTAs on trade in ASEAN. Second, it captures the heterogeneity of the FTA effects by considering imports, exports, total trade, international trade excluding intra-ASEA, international trade including intra-ASEA, and intra-ASEAN trade only. Third, it examines the role of political and economic institutions in shaping the trade creation and diversion effects of the FTAs. In addition, the study accounts for FTA endogeneity, and multilateral resistance terms (MRTs) and used a state-of-the-art estimator, PPML with high-dimensional fixed effect (PPMLHDFE).

Our empirical findings reveal diverse trade creation and diversion effects of FTAs on international and intra-ASEAN trade. And, both political and economic institutions play a significant role in shaping the effects of the FTAs on the trade flows. The findings are extensively discussed in the "Results and Discussion" section and equally summarized in the "Conclusion and Policy Inferences" section of this study.

The remaining part of the article is organized as follows: the "Literature Review" section presents a brief literature review while the "Data and Methods" section deals with the data and methodology. The "Results and Discussion" section contains the analysis and findings, whereas the conclusion is presented in the "Conclusion and Policy Inferences" section.

Literature Review

Over the years, there has been an exponential increase in the number of preferential free trade agreements (FTAs) from about 20 in 1990 to about 300 in 2019. This aroused the interest of researchers and policymakers to find out the reasons for the proliferation of FTAs. Extant literature submits two major reasons. The failure of multilateral trade negotiation. For instance, Rose (2004a) examines the impact of multilateral trade agreements (WTO, GATT, and Generalized System Preferences (GSP)) on the international trade of 175 countries over 50 years. Using the gravity model, the study concludes that there is no evidence that WTO/GATT members have different trade patterns from outsiders (non-members) but GSP has a strong

effect. Rose (2004b) also argues that membership of these organizations does not have a significant effect on trade policy liberalization. The second reason that pushed more countries to sign more preferential free trade agreements is the fear of the domino effect of the multilateral FTA (Baier et al., 2014). This is the fear of being excluded from the existing FTA.

Furthermore, the effect of FTA has been examined by numerous scholars. While the view about the positive effect of FTAs on member countries is less contentious, Viner (1950) argues that FTA has potential trade creation and diversion effects. Therefore, recent studies focus on the heterogoneous and ambiguous welfare effect of the FTA. For instance, Missios et al. (2016) in a game theoretic approach show that custom union agreements prevent Nash equilibrium. That is, membership in trade agreements creates incentives for trade diversion and propelled the non-member countries to lower their tariffs.

Baier et al. (2019) found heterogeneity in the effects of free trade agreements which mostly occur within the FTA rather than across different FTAs. The study also expresses that the effects of the FTAs differ concerning imports and exports depending on the ability of each country to influence the terms of trade of the other. In addition, the effect of new trade agreements is found to be weaker for countries with existing trade agreements. Saucier and Rana (2017) confirm the heterogeneity hypothesis of the effect of preferential trade agreements on international trade. The study submits that labor mobility and environmental provision-based trade agreements promote international trade while capital mobility and competition policy agreements do not have a significant impact on trade. Campi and Dueñas (2019) explore the effect of trade agreements with and without intellectual property rights (IP) on the international trade of 110 countries across the world. The study reports that both trade agreements with and without intellectual property rights influence trade but the effect of the former is stronger than that of the latter. However, the effects become similar when time lag is allowed showing that the implementation of the trade agreements with intellectual property rights requires a longer time than the agreements without intellectual property rights to influence trade. The study further found that the effect of the trade agreements depends largely on the level of development of the countries and the IP intensity of the products traded. In addition, the study submits those trade agreements have an eminent positive effect on trade for the developed countries but there were no important gains for the developing countries. Tabakis and Zanardi (2019) considered antidumping protection and preferential trade agreements in the 15 most intense users of antidumping policy. The study submits that preferential trade agreements weaken (reduce) antidumping protection measures against non-member countries and thus serve as a building block for multilateral trade liberalization.

In the ASEAN region, Okabe and Urata (2014) examine the effect of the ASEAN free trade area (AFTA) agreement on intra-ASEAN trade using the gravity model estimated by the Hausman-Taylor instrumental variable technique. The study finds a positive trade creation effect of the tariff reduction and revealed that the elasticity of tariff reduction on imports is much larger than that of exports. There could be a redirection of trade attention to new trade agreement partners at the expense of existing FTA partners. For instance, Russ and Swanson (2019) evaluated the Korea-U.S free trade agreement

and confirms a strong trade diversion effect on U.S consumption goods imports, particularly from other trading partners having a trade agreement with the USA. In other words, the study reveals that the USA diverted imports away from existing FTA partners toward South Korea. Doan and Xing (2018) observed that membership in ASEAN free trade area improves the efficiency of Vietnam's trade but complex rules of origin and non-member EU and NAFTA hamper Vietnam's trade.

Remarkably, the ability of one country to influence the terms of trade of another country could largely depend on the institutional framework of the latter. Countries with strong political and economic institutions tend to influence the decision of those with weak institutions. This might reshape the trade creation and diversion effects of FTA. However, the extant literature scarcely accounts for the quality of institutions of the trading countries in examining the effect of FTAs. This study tends to fill the research gap by examining the role of political and economic institutions on trade diversion and the trade creation effect of free trade agreements in ASEAN.

Data and Methods

Theoretical Framework

The Anderson and Van Wincoop (2003) structural gravity model consistent with the Helpman and Krupman (1985) model is adopted in this study. The reasons for the choice of the structural gravity model hinges on its numerous desirable properties. It is intuitive, consistent with theory, represents a realistic general equilibrium environment, is flexible in structure, and has strong predictive power (Head & Mayer, 2014; Larch & Yotov, 2016; Yotov et al., 2016). Hence, the structural gravity model is appropriate for quantifying the effect of trade policy such as the trade creation and diversion effects of trade agreement considered. The model expresses that each country involved in bilateral trade produces a certain quantity of a specific bundle of goods. The utility function of a typical consumer, from the demand side, in the destination (importer), is assumed to be the constant elasticity of substitution (CES) type. The consumer's utility maximization problem can be expressed as follows:

$$Maximize\ U\left(X_{ij,t}\right) = \left(\sum_i \beta_{it}^{\frac{1-\sigma}{\sigma}} X_{ij,t}^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}} \tag{1}$$

$$subject\ to\ \sum_i p_{ij,t} X_{ij,t} = y_{j,t} \tag{2}$$

where $\beta_{it}^{1-\sigma/\sigma}$ denotes the number of goods in a unique bundle produced by the exporter (country i) at a given time t. The elasticity of substitution is represented by σ which is assumed to be greater than zero. This implies that the consumers in country j (importer) have a preference to consume the biggest possible number of varieties of goods. The import price from the exporter (country i) and the nominal income of the importer (country j) at time t is denoted by $p_{ij,t}$ and $y_{j,t}$ respectively.

Since international trade involves the cost of shipment of the goods from country i to country j ("iceberg" cost), the country i's producer price of the goods ($p_{i,t}$) at time t is augmented by the shipping cost ($t_{ij,t}$), so that $p_{ij,t} = t_{ij,t}p_{i,t}$. The trade cost (amount lost to the shipping of the goods from country i to country j) is $t_{it,j} - 1$. Therefore, the solution to the optimization problem yielded an import demand equation expressed below.

$$X_{ij,t} = \left(\frac{\beta_{it}p_{it}t_{ij,t}}{p_{j,t}}\right)^{1-\sigma} y_{j,t} \tag{3}$$

where $p_{j,t}$, expressed as follows, represents the importer's ideal price index at a given time t.

$$p_{j,t} = \left(\sum_i (\beta_{i,t}p_{i,t}t_{ij,t})^{1-\sigma}\right)^{\frac{1}{1-\sigma}} \tag{4}$$

Consequently, imposing market clearing conditions, $y_{i,t} = \sum_j X_{ij,t}$, Anderson and Van Wincoop (2003) show that the bilateral trade flow between the importer and exporter can be expressed as follows:

$$X_{ij,t} = \frac{y_{i,t}y_{j,t}}{y_{w,t}}\left(\frac{t_{ij,t}}{p_{i,t}\rho_{j,t}}\right)^{1-\sigma} \tag{5}$$

$$p_{i,t}^{1-\sigma} = \sum\left(\frac{\tau_{ij}}{\rho_j}\right)^{1-\sigma}\frac{y_{j,t}}{y_{w,t}} \tag{6}$$

$$\rho_{j,t}^{1-\sigma} = \sum\left(\frac{t_{ij}}{p_{i,t}}\right)^{1-\sigma}\frac{y_{i,t}}{y_{w,t}} \tag{7}$$

where.

$X_{ij,t}$ = nominal value of exports from country i to country j at time t.

$y_{i,t}$ = nominal GDP of country i (exporter), $y_{w,t}$ = World GDP.

$y_{j,t}$ = expenditure of country j (importer), $t_{ij,t}$ = trade cost (trade barriers).

$p_{i,t}$ = outward multilateral resistance term, $\rho_{j,t}$ = inward multilateral resistance term.

The model insightfully identified that it is the relative prices (not the absolute level of trade barriers) that matter for the size of the bilateral trade flow. The price indices, $p_{i,t}$ and $p_{j,t}$ (multilateral resistance terms), are unobserved and needed to be accounted for in the estimation of the structural gravity model. Estimation of the gravity model without properly accounting for the multilateral resistance terms (MRTs) produces biased and unreliable estimates (Anderson & Van Wincoop, 2003). Some procedures were developed to account for the MRTs. Anderson and Van Wincoop (2003) developed a multi-step nonlinear least square procedure to estimate the MRTs. Baier and Bergstrand (2009) approximated the MRTs by the

"remoteness index" calculated from the GDPs and bilateral distance. However, the Anderson and Van Wincoop (2003) procedure suffers computational complexity and the Baier and Bergstrand (2009) "remoteness index" does not represent the theoretical counterpart of the MRTs (Head & Mayer, 2014). Hummels (2001) and Feenstra (2016) advocate the use of importer-time and exporter-time fixed effects to account for the MRTs. This is adopted in this study because of its advantage in overcoming the computational complexity of Anderson and Van Wincoop's (2003) procedure and as well properly accounting for the MRTs regardless of the kind of data (Olivero & Yotov, 2012).

Empirical Model

Following Dai et al. (2014) and Lach et al. (2019), the empirical model consistent with Anderson and Van Wincoop's (2003) model derived from the Helpman-Krupman (1985) model is specified as follows:

$$X_{ij,t} = \exp\left(\beta_0 + \theta_{ij} + \beta_1 FTA_{ij,t} + \beta_2 FTA_{exp}^{out} + \beta_3 FTA_{imp}^{out} + INST_{i,t} + d_{i,t} + d_{j,t}\right) + \varepsilon_{ij,t} \tag{8}$$

$X_{ij,t}$ represents the bilateral trade flows (exports, imports, and total trade), θ_{ij} denotes the pair-fixed effect which absorbs the time-fixed basic gravity variables (distance, contiguity, common language, and colonial link), serves as a better measure of trade cost, and accounts for the endogeneity of the FTAs (see Baier & Bergstrand, 2007; Agnosteva et al., 2014; Egger & Nigai, 2015). The bilateral trade dummy is denoted by $FTA_{ij,t}$ which assumes one if the exporter (i) and the importer (j) are members of the same trade agreement and zero otherwise. The exporter trade creation/diversion dummy is indicated by FTA_{exp}^{out}. It takes value 1 if the exporter i signs free trade agreement with any partner other than importer j and value 0 otherwise. Similarly, the importer trade creation/diversion dummy is denoted by FTA_{imp}^{out} takes value 1 if importer j signs free trade agreement with any partner other than exporter i and value 0 otherwise. The institutional quality indicators are symbolized by $INST_{i,t}$ while $d_{i,t}$, $d_{j,t}$, and $\varepsilon_{ij,t}$ represent time-varying exporter fixed effect (outward multilateral resistance), time-varying importer fixed effect (inward multilateral resistance), and the stochastic error term respectively.

Equation (8) is estimated using PPML with a high-dimensional fixed effect (PPMLHDFE) estimator developed by Larch et al. (2019). This estimator allows multidimensional fixed effects and the concurrent inclusion of the MRTs and country-specific characteristics such as institutional quality indicators. It equally solves the problems of zero trade flows, heteroscedasticity, endogeneity, and controls for heterogeneity (Correia et al., 2019; Larch et al., 2019; Santos Silva & Tenreyro, 2010).

Data

The study used panel data of ten ASEAN economies and 201 of their trading partners over the period 2000–2018. The choice of the period is based on the

availability of data while the ASEAN countries are chosen due to the strategic position they occupied in world trade. Only the partners that trade with at least six of the ASEAN members over the period considered are included in the sample. The choice of the period is due to the significant changes in the pattern of trade and the proliferation of trade agreements observed across the world during the period. But, 4 years is used for the analysis. The intervals are chosen to control for the gradual adjustment of trade flows in response to trade policy changes and institutional dynamics (Anderson & Yotov, 2016; Baier & Bergstrand, 2007; Cheng & Wall, 2005; Trefler, 2004). In addition, parameter estimates of the trade cost and elasticities obtained from a gravity model with data pooled over consecutive years are usually spurious (Olivero & Yotov, 2012).

The bilateral trade data are obtained from the IMF Direction of Trade Statistics (DOT) while the data on trade agreements are sourced from Mario Larch's regional trade agreements database from Egger and Larch (2008). The indicators of political and economic institutions are taken from World Bank governance indicators and Heritage Foundation respectively. This study used the overall index of economic freedom composed of the unweighted average of the indexes of 12 indicators of economic freedom (property rights, business freedom, trade freedom, financial freedom, government spending, tax burden, fiscal health, freedom from corruption, labor freedom, investment freedom, monetary freedom, and fiscal freedom). The index is measured on a scale of 0–100 with higher values denoting better economic institutions. It is widely accepted as a comprehensive measure of economic institutions methodologically consistent database for international comparison of institutional development (Buterin et al., 2017). In addition, the study used data on the quality of political institutions gotten from the World Bank's Worldwide Governance Indicators (WGI). The WGI consists of six indicators of governance (Kaufmann et al., 2011). Due to the absence of an overall index of political institutions, the principal component analysis (PCA) was used to compute an index of the political institution from the six indicators. This allows for comprehensive coverage of the impact of the political institution on trade diversion or the creation effect of FTA.

Results and Discussion

Effect of FTA and Institutions on International Trade Excluding Intra-ASEAN Trade

The analysis begins with the estimations of the effect of the FTA on trade flows (total trade, exports, and imports) excluding intra-ASEAN trade. The pair-fixed effects are included to adequately capture the effects of standard gravity variables and trade cost and address the issue of endogeneity of trade policy variables (Agnosteva et al., 2014; Baier & Bergstrand, 2007; Egger & Nigai, 2015). The importer-time and exporter-time fixed effect dummies are included in the estimation to account for the exporter (origin) and importer (destination) MRTs. Three dummies (the overall FTA, the exporter-outsider FTA, and the importer-outsider FTA) are included to aptly capture the trade creation or diversion effects of the FTA. Positive and negative

coefficients of the exporter-outsider FTA and the importer-outsider FTA imply trade creation and diversion respectively.

The results of the PPMLHDFE estimates without the indicators of institutional quality are presented in Table 1, columns 1, 2, and 3 for the total trade, exports, and imports respectively. The results show that the FTA has a negative but insignificant effect on trade flows. This indicates that signing numerous FTAs tends to reduce trade with the FTA partners of the ASEAN members. This seems contrary to the a priori expectation. However, it confirms the "spaghetti (noodle) bowl effect" which implies that multiple FTAs slow down trade between FTA member countries (Bhagwati, 1995). This result is reinforced by the positive coefficients of the exporter-outsider FTA which indicates trade creation effects for total trade and exports. Averagely, exporter-outsider FTA leads to 101.6% $[(|e^{0.701} - 1|) \times 100]$ and 121.9% $[(|e^{0.797} - 1|) \times 100]$ trade creation (more trade with non-FTA partners) on total trade and exports respectively. However, the negative coefficient of the exporter-outsider FTA indicates a trade diversion effect on imports. It indicates that exporter-outsider FTA brings about 27.2% $[(|e^{-0.317} - 1|) \times 100]$ reduction in imports with non-FTA partners over the period.

Therefore, the results reveal that the ASEAN economies substitute higher-cost FTA partners with relatively lower-cost non-FTA partners for exports and total trade. This is probably because the ASEAN members are small economies and take advantage of the FTA to expand their markets to outside partners. The FTA provides an opportunity for extending intensive and extensive margins of exports and consequently expanding their trade with all partners including the non-FTA partners. This collaborates with the findings of Doan and Xing (2018) who submits that FTAs positively affect the intensive and extensive margins of imports and exports of Asian countries. This emanates from the voluntary lowering of tariffs by non-members partners (Anderson & Yotov, 2016).

When the indicator of political institutions is included in the model, the result is presented in columns 4, 5, and 6 of Table 1. It shows that political institutions have a negative impact on imports and total trade but positively affects exports. This implies that good political institutions provide an opportunity for the expansion of the export market and reduce the bilateral imports and consequently decrease the total trade. This is because the ASEAN countries import more than they export to their partners. More so, the majority of the ASEAN trade partners (Europe and America) have better political institutions and take advantage of weaker political institutions in the ASEAN countries. So, as ASEAN political institutions become stronger, most of the partners tend to reduce their trade with the ASEAN members. In addition, with political institutions considered, the trade creation effect slightly increased to 103.8% on total trade and reduced to 120.8% on exports while the trade diversion effect on imports declined to 26.8%. The ASEAN countries import and export more from and to their FTA partners respectively but at the same time, the overall trade with non-FTA partners tends to increase as well. This connotes more trade with the rest of the world.

On the other hand, economic institutions have a positive association with imports and total trade but have a negative relationship with exports. Moreover, compared to the model without institutions, the inclusion of the economic institutions leads

Table 1 PPMLHDFE estimates of the effect of FTA and institutions on international trade excluding intra-ASEAN trade

Variables	Without institutions			With political institutions			With Economic institutions		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Trade	Exports	Imports	Trade	Exports	Imports	Trade	Exports	Imports
Free trade agreement (FTA)	−0.0550 (0.0415)	−0.0367 (0.0250)	−0.0713 (0.0592)	−0.0553 (0.0374)	−0.0366 (0.0296)	−0.0715 (0.0460)	−0.0551 (0.0773)	−0.0366 (0.0384)	−0.0714 (0.0647)
Exporter-outsider FTA	0.701*** (0.0508)	0.797*** (0.105)	−0.317*** (0.0658)	0.712*** (0.244)	0.792** (0.333)	−0.312*** (0.103)	0.695*** (0.198)	0.799*** (0.0411)	−0.367** (0.144)
Importer-outsider FTA	−0.0487 (0.0422)	−0.0445 (0.0643)	−0.0615 (0.0656)	−0.0476 (0.0601)	−0.0446 (0.0743)	−0.0574 (0.0899)	−0.0494 (0.106)	−0.0437 (0.0759)	−0.0652 (0.0791)
Political institutions				−0.838*** (0.280)	0.334*** (0.0805)	−1.742*** (0.303)			
Economic institutions							0.0169*** (0.00647)	−0.0219** (0.00921)	0.0830*** (0.0121)
Constant	1.464*** (0.0492)	0.653*** (0.0756)	1.980*** (0.0531)	2.211*** (0.348)	0.351 (0.347)	3.514*** (0.312)	0.241 (0.279)	2.247*** (0.693)	−3.990*** (0.751)
Observations	10,986	13,304	13,298	10,986	13,304	13,298	11,209	13,300	13,452
Exporter-time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Importer-time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wald statistics	221.70	280.41	46.26	19.81	42,305.25	4078.34	456,301.12	1.10e+07	366,301.97
Probability of Wald statistics	0.0000	0.0000	0.0000	0.0005	0.0000	0.0000	0.0000	0.0000	0.0000
Pseudo-*R*-square	0.8402	0.7938	0.8217	0.8402	0.7938	0.8217	0.8409	0.7938	0.8224

Source: Authors' computation

Robust standard errors in parentheses ***, **, and * denote 1%, 5%, and 10% level of significance respectively

to a decrease in the trade creation effect on total trade to 100.4% but an increase in trade creation effect on exports and trade diversion effect on imports to 122.3% and 30.7% respectively. The reason for this is not farfetched. The improvement of economic institutions brings about a rise in income per capita (Acemoglu et al., 2001; Acemoglu et al., 2012). The rise in income might bring about reinforcement of imports concentration as well as imports and export margins across trade partners. Concurrently, the improvement in income increases domestic production and consumption. The excess domestic consumption is augmented by the increase in imports while the concomitant increase in domestic production may lead to expansion in exports to non-FTA partners (see Helpman & Krugman, 1985; Helpman et al., 2008; Masood, 2019). Succinctly, improvement in economic institutions allows the ASEAN members to import more from FTA member partners and export more to non-FTA partners.

Lastly, the results reveal that political institutions have a greater effect on international trade flows than economic institutions. This could be because the economic institutions are endogeneously determined through the collective choices of the society. However, due to conflict of interest over the choice of the economic institutions by various individuals and groups, political power becomes the ultimate arbiter and thus determines the kind of economic institutions that prevails in the society (Acemoglu et al., 2012; Alhassan & Kilishi, 2019).

Effect of FTA and Institutions on International Trade Including Intra-ASEAN Trade

The results in Table 2 are estimates of the structural gravity model for international trade including intra-ASEAN trade. The inclusion of the intra-ASEAN trade is tantamount to the inclusion of intra-national trade obtainable in Anderson and Yotov (2016) and Baier et al. (2014). The intra-national trade is not included in this study because most of the ASEAN members are small economies and Singapore in particular is a city economy. In such economies, intra-national trade flows are negligible compared to international trade flows. The result is similar to that of the model with the international trade excluding intra-ASEAN trade. In addition, the estimates show that the trade creation effect is more on exports (0.984) than the total trade (0.885). On average, an ASEAN member country i that joined FTA with partners other than j, during the sample period, exported $|e^{0.984} - 1| \times 100 = 167.5\%$ more and traded $|e^{0.885} - 1| \times 100 = 142.3\%$ more with non-FTA partners during the sampled period. Moreover, the coefficient of the exporter-outsider FTA is negative and statistically significant for imports. This implies the import diversion effect of the FTA. It shows that the ASEAN countries divert imports from the outside partners toward their FTA partners. The countries averagely imported $|e^{-0.251} - 1| \times 100 = 22.2\%$ less from non-FTA partners during the period considered.

On the other hand, the importer-outsider FTA assumed negative but insignificant coefficients for all trade flows. This implies insignificant trade diversion effects of the importer belonging to outside FTA. The possible explanation for the insignificant effect could be that the ASEAN countries trade with large, open economies such as Europe and America. These partners face a relatively fixed cost of entering

Table 2 PPMLHDFE estimates of the effect of FTA and institutions on international trade including intra-ASEAN trade

Variables	Without institutions			With political institutions			With Economic institutions		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Trade	Exports	Imports	Trade	Exports	Imports	Trade	Exports	Imports
Free trade agreement (FTA)	−0.0458 (0.0308)	−0.0269 (0.0239)	−0.0627 (0.0405)	−0.0457 (0.0308)	−0.0269 (0.0238)	−0.0621 (0.0403)	−0.0455 (0.0308)	−0.0272 (0.0239)	−0.0592 (0.0390)
Exporter-outsider FTA	0.885** (0.354)	0.984*** (0.359)	−0.251* (0.137)	0.887** (0.354)	0.974*** (0.365)	−0.247* (0.135)	0.883** (0.352)	0.987*** (0.358)	−0.277** (0.140)
Importer-outsider FTA	−0.0419 (0.0593)	−0.0124 (0.0769)	−0.0710 (0.0748)	−0.0418 (0.0592)	−0.0129 (0.0765)	−0.0684 (0.0729)	−0.0419 (0.0591)	−0.0122 (0.0770)	−0.0697 (0.0727)
Political institutions				−0.123** (0.0567)	0.753*** (0.0787)	−0.989*** (0.148)			
Economic institutions							0.00884** (0.00355)	−0.0311*** (0.0103)	0.0746*** (0.0170)
Constant	1.385*** (0.332)	0.578* (0.348)	1.996*** (0.109)	1.492*** (0.323)	−0.0972 (0.293)	2.829*** (0.201)	0.746** (0.343)	2.841*** (0.952)	−3.364*** (1.149)
Observations	11,554	13,793	13,771	11,554	13,793	13,771	11,552	13,789	13,767
Exporter-time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Importer-time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Wald statistics	9.49	8.16	22.30	243.27	1,410,438.23	485.21	48.41	107.84	134.93
Probability of Wald statistics	0.0234	0.0428	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Pseudo-*R*-square	0.8501	0.8063	0.8285	0.8501	0.8063	0.8285	0.8501	0.8063	0.8285

Source: Authors' computation

Robust standard errors in parentheses ***, **, and * denote 1%, 5%, and 10% level of significance respectively

the ASEAN market (see Melitz, 2003) and only divert minute components of their trade to satisfy the requirement for the FTA. Such diversion might not significantly affect the trade flows.

When the indicators of political institutions are added to the model, the PPML-HDFE estimates are presented in columns 4, 5, and 6 for total trade, exports, and imports respectively. The results show that the coefficients of the exporter-outsider FTA remain positive and robustly significant with higher values for total trade and lower values for exports and imports. Furthermore, the results show that political institutions have a positive effect on exports but negative effects on imports and total trade in the ASEAN countries. Better political institutions increase exports by 212.3% ($|e^{0.753}|\times100$) but reduces imports and total trade by ($|e^{-0.123}|\times100$) 88.4% and ($|e^{-0.989}|\times100$) 37.2% respectively. The positive coefficient of the political institutions indicates that better governance reduces the trade creation effect on export by increasing exports to the FTA partners. With the inclusion of political institutions into the model, the export to non-FTA partners drops from 167.5 to164.9%. Contrarily, the negative coefficient implies that improvement in political institutions increases the trade creation effect (from 142.2 to 142.8%) on total trade and reduces the trade diversion effect on imports (from 22.2 to 21.89%) by reducing total trade with FTA partners and increasing imports from FTA partners respectively. Thus, the inclusion of the political institutions has a significant impact on the trade creation ad diversion effect of bilateral trade flows in ASEAN. But, the importer-outsider FTA remains statistically insignificant. This is an indication that good political institutions amplify the trade creation effect of the ASEAN countries engaging in outsider FTA agreements, and the engagement of the importers on FTA with other partners rather than the ASEAN members does not significantly affect trade flows. This intuitively implies that strong political institutions neutralize the negative effect of importer trade diversion on non-FTA partners.

Similarly, the results in columns 7, 8, and 9 show that the economic institutions positively affect total trade and imports. Meanwhile, the effect of the political institutions is greater than that of the economic institutions and the effect of both is more on imports than exports and total trade. Averagely, a percentage rise in economic institutions enhances total trade and imports by 100.9% ($|e^{0.00884}|\times100$) and 107.7% ($|e^{0.0746}|\times100$) but reduces exports by 96.9% ($|e^{-0.0311}|\times100$) respectively. The improvement in economic institutions fosters income and promotes import and total trade. However, the rise in imports could hamper domestic production and exports. In addition, the inclusion of the economic institutions lead to a decrease in the total trade creation effect of exporter-outsider FTA from 142.3 ($|e^{-0.0311}|\times100$) to 141.8% ($|e^{0.883}-1|\times100$), increase in trade creation effect on exports from 167.5 to 168.3%, and an increase in import diversion effect of the exporter-outsider FTA from 22.2 to 24.2%. The reason for the greater effect of the political institutions on export is not farfetched. Political institutions are observed to be important determinants of the quality of economic institutions (Acemoglu et al., 2012; Alhassan & Kilishi, 2019). The inclusion of the indicators of institutional quality plays a significant role in changing the trade creation and diversion effects of the FTA on international trade including intra-ASEAN trade.

Effect of FTA and Institutions on Intra-ASEAN Trade

The estimation is restricted to only intra-ASEAN trade and the results are reported in Table 3 where columns 1, 2, and 3 contain the estimates without institutions for total trade, exports, and imports respectively. The results with political and economic institutions are displayed in columns 4–6 and 7–9 respectively. Only the intra-ASEAN FTA and exporter-outsider FTA dummies are included to represent the intra-regional FTA and the trade creation/diversion effects respectively. This is because the trade involves only the ASEAN members and as such importers and exporters are the same countries. The result indicates that the intra-ASEAN FTA has positive coefficients for total trade and export but negative for imports. This implies that bilateral trade agreements among ASEAN members facilitate intra-ASEAN trade. However, the negative and significant coefficients of the intra-ASEAN FTA on imports reflect the noodle bowl (slow down) effect of criss-crossing FTAs on intra-ASEAN imports. This could be a result of the fact that the ASEAN economies preferably import from their outside trade partners rather than themselves. For instance, ASEAN is the third largest trading partner of the EU after the USA and China (European Commission, 2019). So, as trade among themselves intensifies, more is imported from outside partners. Without the effects of institutions, intra-ASEAN FTA improves intra-ASEAN total trade and exports by 17.5% ($|e^{0.161} - 1| \times 100$) and 65.9% ($|e^{0.506} - 1| \times 100$) respectively while it leads to a decline in intra-ASEAN imports by 20.7% ($|e^{-0.232} - 1| \times 100$). The estimates also show a trade creation effect on total trade and exports while the trade diversion effect is recorded for imports. This is indicated by the positive coefficients of exporter-outsider FTA for total trade and exports and the negative coefficients for imports in all the models (with and without institutions). That is, outsider FTA among the ASEAN countries diverts imports from non-ASEAN FTA partners and creates exports. This implies importing less generally and exporting and trading more with the rest of the world. In short, multiple FTAs among the ASEAN countries is baneful for imports between the countries but beneficial for overall trade and exports. The tenable explanation for this is that, though regional trade agreements may be beneficial, multiple intra-ASEAN FTAs could restrict intensive and extensive margins of imports which leads to import diversion and a noodle bowl effect on imports.

The estimates in column 4, 5, and 6 of Table 3 represent the results of the models which includes the indicators of political institutions. The results show that political institutions positively affect exports and negatively affect imports but are insignificant for the total trade. Meaning, an improvement in the political institutions of the ASEAN members brings about an increase in intra-ASEAN exports and a decline in imports. This suggests that ASEAN economies with good institutions export more and import less among themselves. Moreover, the inclusion of the political institutions slightly reduces the trade creation and diversion effect by enhancing trade with non-ASEAN countries. That is more trade with the rest of the world. This is because most of the trading partners of the ASEAN countries have better political institutions. Hence, improvement in institutional proximity enhances trade flows between the ASEAN members and the rest of their partners outside ASEAN.

Table 3 PPMLHDFE estimates of the effect of FTA and institutions on intra-ASEAN

Variables	Without institutions			With political institutions			With Economic institutions		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Trade	Exports	Imports	Trade	Exports	Imports	Trade	Exports	Imports
Intra-ASEAN free trade agreement (FTA)	0.161*** (0.0283)	0.506*** (0.0833)	−0.232*** (0.00182)	0.160*** (0.0403)	0.498*** (0.0898)	−0.218*** (0.00262)	0.837** (0.368)	1.055*** (0.185)	0.00276 (0.268)
Exporter-outsider FTA	0.180*** (0.0263)	0.633*** (0.0473)	−0.286*** (0.00454)	0.177*** (0.0434)	0.624*** (0.0449)	−0.270*** (0.0171)	0.853** (0.375)	1.179*** (0.329)	−0.0512 (0.192)
				0.0616 (0.159)	0.272*** (0.00358)	−0.477** (0.234)			
Political institutions							0.265*** (0.0878)	0.260*** (0.0469)	0.0811 (0.0920)
Constant	2.751*** (0.00712)	1.958*** (0.0128)	2.250*** (0.00525)	2.705*** (0.109)	1.726*** (0.0153)	2.557*** (0.149)	−16.45** (6.437)	−17.18*** (3.537)	−3.541 (6.540)
Observations	343	356	356	343	356	356	343	356	356
Exporter/importer-time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo-*R*-square	0.8508	0.8039	0.7932	0.8508	0.8039	0.7932	0.8509	0.8040	0.7932

Source: Authors' computation

Robust standard errors in parentheses ***, **, and * denote 1%, 5%, and 10% level of significance respectively

Similarly, the results in columns 7, 8, and 9 show that economic institutions have a positive relationship with intra-ASEAN exports and total trade but are insignificant for the imports. However, the inclusion of the economic institutions in the model led to an enormous increase in the positive effect of intra-ASEAN FTA as well as exporter-outsider FTA trade creation effect on exports and total trade while the trade diversion effect on imports becomes statistically insignificant. In other words, improvement in economic institutions brings about an expansion in trade within and outside ASEAN. It also enhances the trade-promoting effect of the intra-ASEAN FTA. In a nutshell, both political and economic institutions significantly affect the dynamics of the effect of the free trade agreement on trade flows among ASEAN countries. But, good political institutions facilitate the trade at less than a proportionate improvement which results from good economic institutions.

Conclusion and Policy Inferences

The main objective of this study is to evaluate the role of economic and political institutions concerning the effect of FTA on trade in ASEAN. We considered the trade diversion and creation effects of the FTAs and captures the heterogeneity of the FTA effects by considering imports, exports, total trade, international trade excluding intra-ASEA, international trade including intra-ASEA, and intra-ASEAN trade only. We also examine the role of political and economic institutions in shaping the trade creation and diversion effects of the FTAs. In addition, the study accounts for FTA endogeneity and multilateral resistance terms by estimating a structural gravity model with the state-of-the-art estimator, PPML with high-dimensional fixed effect (PPMLHDFE).

We find a trade creation effect on international exports and total trade but diversion on imports resulting from exporter-outsider FTA. However, the trade creation effect is more on export than total trade. On the other hand, there is an insignificant importer-outsider FTA and spaghetti bowl effect of FTAs on international trade flows. The study further reveals that political institutions have a positive association with international exports but negatively affect imports and total trade. Meanwhile, the inclusion of political institutions amplifies exporter-outsider FTA total trade creation effect and reduces export creation and import diversion. This implies that with the inclusion of political institutions, overall trade with non-FTA partners increases.

Similarly, economic institutions have a positive relationship with international imports and total trade but a negative relationship with exports. In addition, trade creation on total trade decreases while that of export creation and import diversion increases when the indicators of economic institutions are captured in the model. In other words, political and economic institutions affect trade creation/diversion effects in opposing directions. But, political institutions have a greater impact than economic institutions on international trade flows while the reverse is the case for intra-ASEAN trade.

Considering intra-ASEAN trade, we find that intra-ASEAN FTA promotes intra-ASEAN exports and total trade but slows down intra-ASEAN imports (noodle bowl problem). With the political institutions considered, trade diversion and creation

effects reduce and intra-ASEAN trade improves with good political institutions. Likewise, strong economic institutions promote intra-ASEAN trade. We further establish that intra-ASEAN trade creation effects on export and total trade increase while trade diversion effect on imports becomes insignificant when economic institutions are included. This shows that improvement in economic institutions brings about an expansion in trade within and outside ASEAN. It also enhances the trade-promoting effect of the FTA. In a nutshell, the inclusion of the indicators of political and economic institutions into the specification significantly changed the magnitude of the trade creation and diversion effects of FTA on international and intra-ASEAN trade. And, the effects of the quality of institutions are robust to the different trade flows (total trade, exports, and imports) of the ASEAN economies.

Data Availability Data will be provided on request.

Declarations

Conflict of Interest The authors declare no competing interests.

References

Acemoglu, D., Johnson, S., & Robinson, J. A. (2001). The colonial origins of comparative development: An empirical investigation. *American Economic Review, 91*(5), 1369–1401.

Acemoglu, D., Johnson, S., & Robinson, J. A. (2012). The colonial origins of comparative development: An empirical investigation: Reply. *American Economic Review, 102*(6), 3077–3110.

Agnosteva, D. E., Anderson, J. E., & Yotov, Y. V. (2014). Intra-national trade costs: Measurement and aggregation (No. w19872). *National Bureau of Economic Research*.

Alhassan, A., & Kilishi, A. A. (2019). Weak economic institutions in Africa: A destiny or design? *International Journal of Social Economics, 46*(7), 904–919.

Alhassan, A., & Payaslioglu, C. (2019). Institutions and bilateral trade in Africa: An application of Poisson's estimation with high-dimensional fixed effects to structural gravity model. *Applied Economics Letters*, 1–5. https://doi.org/10.1080/13504851.2019.1682112

Anderson, J. E., & Van Wincoop, E. (2003). Gravity with gravitas: A solution to the border puzzle. *American Economic Review, 93*(1), 170–192.

Anderson, J. E., & Yotov, Y. V. (2016). Terms of trade and global efficiency effects of free trade agreements, 1990–2002. *Journal of International Economics, 99*, 279–298.

Aziz, N., Hossain, B., & Mowlah, I. (2018). Does the quality of political institutions affect intra-industry trade within trade blocs? *The ASEAN Perspective. Applied Economics, 50*(33), 3560–3574.

Baier, S. L., & Bergstrand, J. H. (2007). Do free trade agreements actually increase members' international trade? *Journal of International Economics, 71*(1), 72–95.

Baier, S. L., & Bergstrand, J. H. (2009). Estimating the effects of free trade agreements on international trade flows using matching econometrics. *Journal of International Economics, 77*(1), 63–76.

Baier, S. L., Bergstrand, J. H., & Mariutto, R. (2014). Economic determinants of free trade agreements revisited: Distinguishing sources of interdependence. *Review of International Economics, 22*(1), 31–58.

Baier, S. L., Yotov, Y. V., & Zylkin, T. (2019). On the widely differing effects of free trade agreements: Lessons from twenty years of trade integration. *Journal of International Economics, 116*, 206–226.

Belloc, M. (2006). Institutions and international trade: A reconsideration of comparative advantage. *Journal of Economic Surveys, 20*(1), 3–26.

Bhagwati, J. N. (1995). US Trade Policy: The infatuation with free trade areas. In J. Bhagwati & A. O. Krueger (Eds.), *The dangerous drift to preferential trade agreements* (pp. 1–18). The AEI Press.

Buterin, V., Škare, M., & Buterin, D. (2017). Macroeconomic model of institutional reforms' influence on economic growth of the new EU members and the Republic of Croatia. *Economic Research-Ekonomska Istraživanja, 30*(1), 1572–1593.

Campi, M., & Dueñas, M. (2019). Intellectual property rights, trade agreements, and international trade. *Research Policy, 48*(3), 531–545.

Cheng, I. H., & Wall, H. J. (2005). Controlling for heterogeneity in gravity models of trade and integration. *Federal Reserve Bank of St. Louis Review*, (Jan), 49–63.

Correia, S., Guimarães, P., & Zylkin, T. (2019). PPMLHDFE: Fast poison estimation with high-dimensional fixed effects. *arXiv preprint* arXiv:1903.01690.

Dai, M., Yotov, Y. V., & Zylkin, T. (2014). On the trade-diversion effects of free trade agreements. *Economics Letters, 122*(2), 321–325.

Doan, T. N., & Xing, Y. (2018). Trade efficiency, free trade agreements and rules of origin. *Journal of Asian Economics, 55*, 33–41.

Egger, P. H., & Nigai, S. (2015). Structural gravity with dummies only: Constrained ANOVA-type estimation of gravity models. *Journal of International Economics, 97*(1), 86–99.

Egger, P., & Larch, M. (2008). Interdependent preferential trade agreement memberships: An empirical analysis. *Journal of International Economics, 76*(2), 384–399.

European Commission. (2019). EU trade agreements implementation annual report. retrieved from https://ec.europa.eu/commission/presscorner/detail/en/ip_19_6074

Feenstra, R. C. (2016). *Advanced international trade: Theory and evidence*. Princeton university press.

Head, K., & Mayer, T. (2014). Gravity equations: Workhorse, toolkit, and cookbook. In *Handbook of international economics*, *4*, 131–195. Elsevier.

Helpman, E., & Krugman, P. R. (1985). *Market structure and foreign trade: Increasing returns, imperfect competition, and the international economy*. MIT press.

Helpman, E., Melitz, M., & Rubinstein, Y. (2008). Estimating trade flows: Trading partners and trading volumes. *The Quarterly Journal of Economics, 123*(2), 441–487.

Hummels, D. L. (2001). Toward a geography of trade costs. *Available at SSRN 160533*.

Kaufmann, D., Kraay, A., & Mastruzzi, M. (2011). The worldwide governance indicators: Methodology and analytical issues. *Hague Journal on the Rule of Law, 3*(2), 220–246.

Larch, M., & Yotov, Y. V. (2016). General equilibrium trade policy analysis with structural gravity.

Larch, M., Wanner, J., Yotov, Y. V., & Zylkin, T. (2019). Currency unions and trade: A PPML re-assessment with high-dimensional fixed effects. *Oxford Bulletin of Economics and Statistics, 81*(3), 487–510.

Lee, J. W., & Shin, K. (2006). Does regionalism lead to more global trade integration in East Asia? *The North American Journal of Economics and Finance, 17*(3), 283–301.

Masood, M. (2019). New evidence on income and the geographical distribution of imports: The case of audiovisuals. *Journal of Comparative Economics*.

Melitz, M. J. (2003). The impact of trade on intra-industry reallocations and aggregate industry productivity. *econometrica*, *71*(6), 1695–1725.

Missios, P., Saggi, K., & Yildiz, H. M. (2016). External trade diversion, exclusion incentives and the nature of preferential trade agreements☆. *Journal of International Economics, 99*, 105–119.

Okabe, M., & Urata, S. (2014). The impact of AFTA on intra-AFTA trade. *Journal of Asian Economics, 35*, 12–31.

Olivero, M. P., & Yotov, Y. V. (2012). Dynamic gravity: Endogenous country size and asset accumulation. *Canadian Journal of Economics/revue Canadienne D'économique, 45*(1), 64–92.

Osabuohien, E. S., Beecroft, I., & Efobi, U. R. (2018). Global trade and trade protection in a globalised world. *Transnational Corporations Review, 10*(1), 43–52.

Rose, A. K. (2004a). Do we really know that the WTO increases trade? *American Economic Review, 94*(1), 98–114.

Rose, A. K. (2004b). Do WTO members have more liberal trade policy? *Journal of International Economics, 63*(2), 209–235.

Russ, K. N., & Swenson, D. L. (2019). Trade diversion and trade deficits: The case of the Korea-US free trade agreement. *Journal of the Japanese and International Economies, 52*, 22–31.

Santos Silva, J., & Tenreyro, S. (2010). On the existence of the maximum likelihood estimates in Poisson regression. *Economics Letters, 107*(2), 310–312.

Saucier, P., & Rana, A. T. (2017). Do preferential trade agreements contribute to the development of trade? Taking into account the institutional heterogeneity. *International Economics, 149*, 41–56.

Schüle, U., & Kleisinger, T. (2016). *The "Spaghetti Bowl": A case study on processing rules of origin and rules of cumulation* (No. 2/2016). UASM Discussion Paper Series.

Soeng, R., & Cuyvers, L. (2018). Domestic institutions and export performance: Evidence for Cambodia. *The Journal of International Trade & Economic Development, 27*(4), 389–408.

Tabakis, C., & Zanardi, M. (2019). Preferential trade agreements and antidumping protection. *Journal of International Economics, 121*, 103246.

Trefler, D. (2004). The long and short of the Canada-US free trade agreement. *American Economic Review, 94*(4), 870–895.

Viner, J. (1950). The customs union issue (Carnegie Endowment for Peace). *Viner the Customs Union Issue 1950*.

World Trade Organization. (2019). WTO Annual Report.

Yotov, Y. V., Piermartini, R., & Larch, M. (2016). *An advanced guide to trade policy analysis: The structural gravity model*. WTO iLibrary.

Zeynalov, A. (2017). The gravity of institutions in a resource-rich country: The case of Azerbaijan. *International Economics and Economic Policy, 14*(2), 239–261.

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